# Simio Project Final Report

IE 4530

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# Problem Statement and Conceptual Model

## Introduction & Background

In this project, the team will create a simulation modeling a department store during the Black Friday holiday. The post-Thanksgiving season sees an annual influx in consumerism, specifically in the retail sector. There is potential for bottlenecks in logistics inside shopping centers. For this project, the model will be based on the Macy's located in Baton Rouge. It will focus on simulating the average customer experience, showing the entire process of entering to exiting the store, with all potential in between processes such as waiting in line, using the fitting rooms, checking out, shopping in departments, balking, and reneging.

The team will structure the simulation model by first fully understanding the layout and operations of the Baton Rouge Macy's. The team will determine worker capacity, average shopping patterns, entrances, and exit points. Average customer arrival times will reflect the expected shopper surges; for example, if doors open at 6:00 AM, there may be shoppers waiting outside for an early bird special, while other customers come in later in the day. This will be expressed using a rate table, which will allow for a pre-set number of customers to come in at given intervals throughout the day. Customers will be represented by entities that will enter the system following various paths connecting to different locations throughout the store. There are different departments within Macy's, including Backstage, Shoes and Juniors, Women's, Men's and Children's, Perfume, Jewelry, and Furniture and Appliances. Servers will be used for a multitude of purposes; cashiers will be able to perform transactions, returns, exchanges, etc. With multiple queues, customers can choose the shortest line.

## Management Objectives

Upper-level management wants to improve the Black Friday experience for customers by streamlining queues and creating more efficient flow. The Black Friday holiday can cause a large influx of visitors at one singular time, which stresses the current system. Besides the issue of

magnitude, customers may enter the store with varying needs. Especially within a department store, there is high variability among the services offered.

By increasing server utilization across all departments, management will hope to see an increase in throughput and decrease in averages for queue length, wait time, and flowtime. Ideally, the number of customers lost due to reneging will also decrease. Overall, these improvements will increase customer satisfaction, return rates, and revenue for the business.

## Problem (or System) Description

The proposed model will closely replicate the standard Macy's department store. For modeling purposes, an image of the store's layout will be used, modeled to scale of the actual dimensions. The number of customers that will enter the store on Black Friday was gathered from discussions with management, and sensitivity analysis will be used to find the true number of purchasing customers the system can handle. The current system operation functions as a typical retail business: customers will be able to browse, utilize fitting rooms, make returns/exchanges, check out, or leave the store without purchasing anything. There will be congestion for nearly all simulated servers for each business, as Black Friday causes an above average influx of customers. The issue lies within the quantity and inconsistent flow of customers and a limited number of resources to accommodate. The primary focus will be on the queue lengths, wait times, server utilization, and flow times.

## Conceptual Model - Existing System

The conceptual diagram, derived from a sample Macy's Visual Merchandising portfolio (sample layout shown in Appendix Figure 1 and 2), models the workflow and the layout used to coordinate operations across more than 450 stores nationwide. Each store has a slightly different physical layout, so simulations will be modeled after the Baton Rouge Macy's store layout, depicted in Figures 1 and 2.

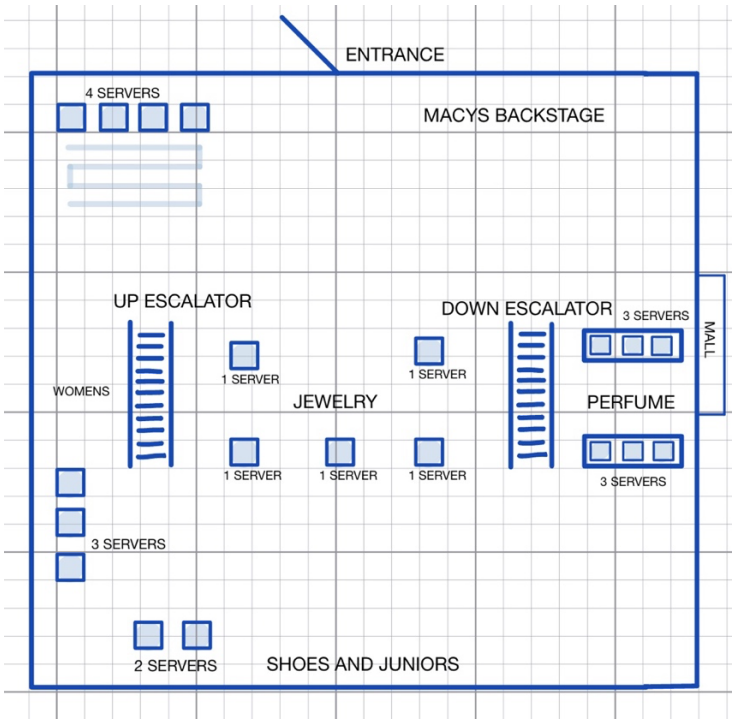



*Figure 1: First Floor Macy's Model*

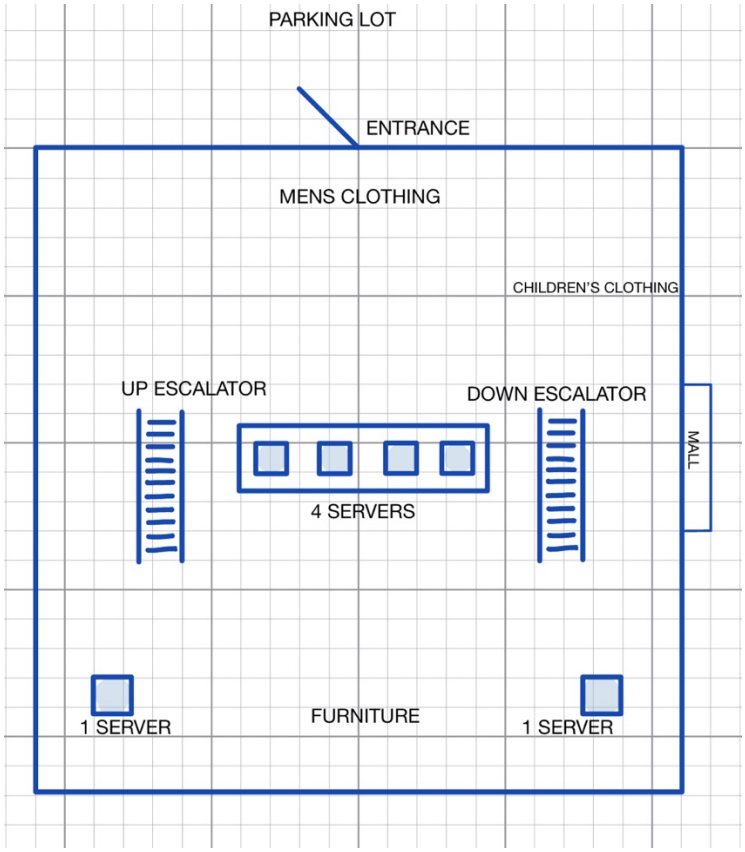



*Figure 2: Second Floor Macy's Model*

Customers will see the Women's section in the same vicinity as the Perfume, Jewelry and Juniors sections; moreover, the Men's will be by the Children's and the home goods area. There are a multitude of servers in each section along with corresponding lines. Customers frequently experience congestion due to long queue lengths and long flow times depending on the people before them.

## Alternative(s)

For the Black Friday simulation project, two alternatives the team is considering using are relocating server capacity and using a constant arrival rate table. The primary alternative to reallocate server capacity across departments is based on an observed demand pattern in the simulation model. After analyzing customer path length, routing logic, and the number of entities moving toward each station, the team identified which departments consistently faced higher customer volume, longer travel distances, and abnormal flowtimes. Using these findings, workers were redistributed or increased so that the sections with heavier demand were assigned additional servers. This lean-based alternative focused on line balancing to reduce bottlenecks, shorten flow times and improve overall system throughput.

The second alternative the team evaluated was implementing a constant arrival rate table to smooth customer entry into the store. The typical surge on this holiday is customers rushing in at opening, creating immediate congestion and overwhelming smaller departments. To address this, the constant arrival rate approach models a controlled release of a fixed number of customers at specific time increments, reducing the initial spike in demand and creating a steady flow throughout the early hours. This strategy aligns with waste reduction principles by decreasing variability, limiting bottleneck formation and improving the overall shopping experience inside the store.

Outside of the simulation model, the team is also considering buying online or pickup in store options, where customers can purchase items online and have several days to pick up their orders, reducing in-store congestion on Black Friday. While this specific alternative cannot directly be modeled for the simulation, it is connected to waste reduction, as it shifts demand away from peak periods and improves the overall shopping experience.

## Evaluation

The alternatives will be evaluated based on their impact on system performance metrics. The specific metrics we will work to achieve include increasing server utilization by 5-10%, which will be accomplished by balancing server capacity to customer need, decreasing average flowtime, and increasing throughput of customers. The central success criterion is that all customers who enter the system successfully complete their shopping trip and exit before store closing, indicating the redesigned system can handle peak Black Friday demand without excessive congestion or unmet services.

## Data Collection

By using statistics from the National Retail Federation, the team was able to collect country-wide data on the total number of adults who participate in Black Friday Shopping, both in person and online (Farrell, 2023). In 2023, 200 million shoppers hit the stores on the five-day Black Friday weekend, with roughly half of them being instore shoppers. These yearly estimates were scaled to the population size of Baton Rouge to create a rough estimate on the number of shoppers at the Mall of Louisiana.

Of all shoppers on Black Friday in 2023, 76.2 million shopped in store. To get a ballpark estimate of the total shoppers the Mall of Louisiana could see, the estimated number of adults in Baton Rouge and the surrounding areas can be totaled and scaled by the percentage of instore shoppers nation-wide (Casey's Kids Count Team, 2024) (Census Reporter, 2023).

$$Total\ possible\ shoppers\ in\ Baton\ Rouge\ (ages\ 18-70): 200k$$

$$Total\ possible\ population\ of\ shoppers\ nation-wide: 220M$$

$$Percentage\ of\ shoppers, compaired\ to\ population: \frac{76.2\ M}{220\ M} = 34.6\%$$

$$Possible\ shoppers\ to\ Mall\ of\ Louisiana: 200{,}000 * 0.346 = 69.2k$$

After calculations, about 69k visitors are estimated to arrive at the Mall of Louisiana on the Black Friday holiday. The team found that around 34.6% of the adult population within the United States shops in store on Black Friday, as calculated from the National Retail Foundation. By using this percentage alongside the total adult population of the Baton Rouge area that utilized the Mall of Louisiana, the team calculated the in-store shopper expectancy.

After speaking with management, the team recorded that the annual number of customers expected by Macy's each Black Friday is around 35,000. When compared to a possible total of 69,000 customers, the store estimates that around half of visitors will enter Macy's. To test the capabilities of the system, ranges of customer totals, centered around 35,000, will be tested.

The last aspect of data collection will involve estimations of interarrival times for customers. Macy's offers unique sales on hot commodity items, such as 60% off bedding and 25% off Nike clothing. As a result of these deals, customers tend to arrive at higher volumes earlier in the day. The team will simulate these arrivals by utilizing an arrival table for customers, so that more customers arrive at opening hours, between 6:00 AM and 9:00 AM. There will then be another peak in visitors during the afternoon from 12:00 PM to 2:00 PM, which is post lunch rush.

# Input Analysis

## Methods for Input Data Analysis

The team visited the Macy's located in the Mall of Louisiana to collect data on the current processing system for customers. A map of the store, shown in Appendix Figure 3 and 4, highlight the total number of cashier stations for the basis of the simulation design. The team spoke with two employees and one manager during the visit.

Both employees mentioned that Black Friday is the busiest time of the year for the store, and queues at checkout will be an average of 20 people long, with each waiting at minimum 10-15 minutes before they are able to be checked out. The reason behind the long waiting times is not only due to high customer arrival rates, but also due to specific purchases. Macy's will place high-demand clothing items, such as the newest winter accessories, and household appliances, at

discounts of 20-50% off. Each customer may be purchasing a large range of products of various sizes, leading to longer processing times during checkout when compared to usual checkout times. The store manager stated that the Baton Rouge location sees 35,000 visitors on Black Friday annually. The store is expecting the same number of visitors this year.

There are a total of 26 service stations throughout the location, each representing a cashier station. Appendix figures 3 and 4 show the location of all the cashier stations. The servers are broken down into specific departments:

- Men's and children's clothing: 4 service stations
- Furniture/household appliances: 2 service stations
- Macy's Backstage: 4 service stations
- Women's clothing: 3 service stations
- Shoes and junior's clothing: 2 service stations
- Jewelry: 5 service stations
- Perfume: 6 service stations

Jewelry, Perfume, and Macy's Backstage servers must be used by customers shopping for goods in those departments. Those departments will not check out goods from other departments. Therefore, there will be no queue jockey (moving from one line to another) in these departments. All cashiers within clothing, shoes, and furniture departments will service any customer shopping within these specific departments. According to management, 30% of visitors will balk before entering the system, meaning that they leave the store without being processed at a server.

## Results of Data Collection

For the customer arrival patterns, the team will use information provided from management to create a rate table. The team is investigating the usage of the exponential function to model arrival patterns, due to management only having data on the average number of arrivals. The exponential distribution family matches the expected data, since customer arrivals per hour are based on an average expected value. Interarrivals are not constant because specific arrivals are based on the individual shopper. Instead, these rates vary, but average to a mean value.

Management noted that arrivals are the most frequent between 8:00 AM to 9:00 AM, and there is another higher arrival period around lunch time, from 12:00 PM to 2:00 PM. Average arrivals per hour were estimated based on management input, and the total arrivals per hour were

calculated as customer arrival rates per minute. Table 1 holds all interarrival times that will be used within Simio for modeling the process. The original will be tested first, followed by decreased totals that match the service capabilities of the system.

*Table 1: Interarrival Times for Model Design*

| Hours | Entities per arrival | 20% Sensitivity Analysis | 30% Sensitivity Analysis | 40% Sensitivity Analysis |
|---|---|---|---|---|
| 6AM-7AM | 1750 | 1400 | 1225 | 1050 |
| 7AM-8AM | 1750 | 1400 | 1225 | 1050 |
| 8AM-9AM | 7000 | 5600 | 4900 | 4200 |
| 9AM-10AM | 2800 | 2240 | 1960 | 1680 |
| 10AM-11AM | 2100 | 1680 | 1470 | 1260 |
| 11AM-12PM | 2100 | 1680 | 1470 | 1260 |
| 12PM-1PM | 3150 | 2520 | 2205 | 1890 |
| 1PM-2PM | 2800 | 2240 | 1960 | 1680 |
| 2PM-3PM | 2450 | 1960 | 1715 | 1470 |
| 3PM-4PM | 1750 | 1400 | 1225 | 1050 |
| 4PM-5PM | 1750 | 1400 | 1225 | 1050 |
| 5PM-6PM | 1750 | 1400 | 1225 | 1050 |
| 6PM-7PM | 1750 | 1400 | 1225 | 1050 |
| 7PM-8PM | 1400 | 1120 | 980 | 840 |
| 8PM-9PM | 350 | 280 | 245 | 210 |
| 9PM-10PM | 350 | 280 | 245 | 210 |
| 10PM-11PM | 0 | 0 | 0 | 0 |
| **Total Customers** | 35000 | 28000 | 24500 | 21000 |

For processing times, the most representative distribution of the variety of times would be triangular distribution. By speaking to the manager, length of transactions depends on the number of items, if there are any manual returns, if the individual is applying for a credit card, and the packaging/preparation of each piece (removing hangers, wrapping delicates, folding items, etc.). According to management, the minimum processing time observed is 1 minute, max time is 3 minutes, and usually the processing time is about 2 minutes. Workers are trained during Black Friday to respond quickly to guests and ensure the quickest transaction possible.

## Problems, Limitations, Biases

While the average customer inflow for Black Friday annually has remained at 35,000 visitors, the possibility of rain or extreme weather can affect the total number of visitors who may come to the store. As the number of online has increased over the past years, from 18.8% in 2021 to currently 21% in 2025 and an expected growth to 22.6% in 2027 (Satistica, 2025), it is clear the eCommerce market is only experiencing an upward trend. While there are not many academic journals studying the relationship between weather and eCommerce (largely due to the difficulty of matching geographical locations and their respective weather conditions), it can be reasonably

assumed that it is more convenient for customers to access online shopping platforms than to travel to the location of the department store, especially when there is heavy rain or other extreme conditions (Li, 2024). Black Friday falls within Louisiana's hurricane season, so this is just another possibility to consider when investigating the validity of the data.

While the team has access to the average amount of arrivals, there is limited data on specifics necessary to map the process. The team will have to use some random data, based on averages provided, due to the lack of data clarity offered by Macy's. However, unknowns will be balanced by sensitivity analysis, as explained previously.

Different departments of Macy's may not experience the same traction as another. Depending on customer needs as well as the value of the discount, customers may find themselves more drawn to one department over another. The team was unable to consult workers in all of the departments, so this is another limitation of the model.

Employees and managers may also be inflating the values, for the reputation of the business or for other reasons. Since the team did not collect data on the actual event of Black Friday, the numbers used are based off personal anecdotes (mixed in with estimates and assumptions) instead of tallying customers.

Since Macy's is a corporate-owned retail location, solutions suggested by the team may not be feasible. Without knowing any constraints, the team is exploring several options that may not be within the company's limitations (unrealistic solutions, etc.).

# Lean Analysis

## Value Stream Map

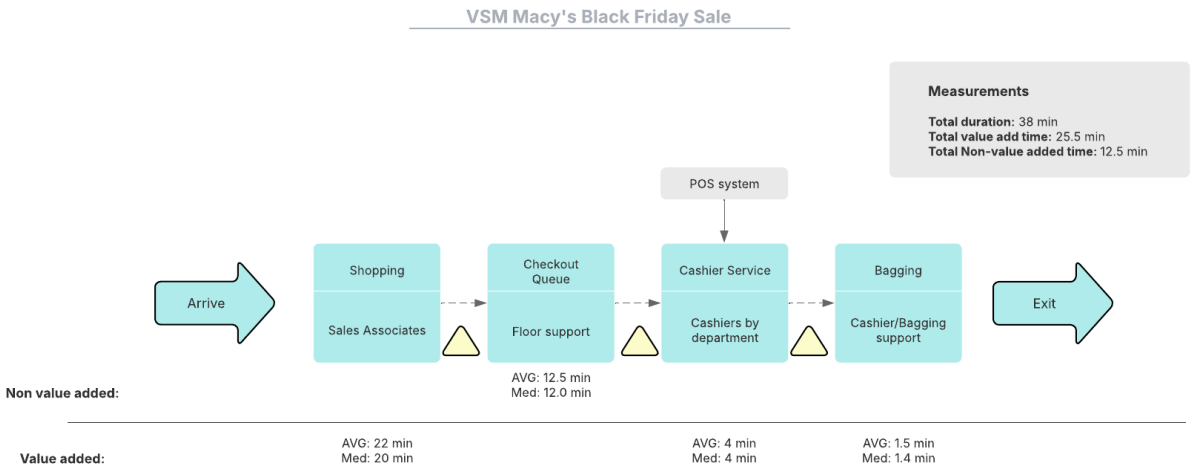



*Figure 3: VSM*

To better understand and improve the customer flow experience during Black Friday at Macy's, a current-state Value Stream Map (VSM) was developed (Figure 3). This VSM visualizes each stage of the customer journey, from arrival, shopping, and waiting in the checkout queue, through cashier processing, bagging, and exit, and quantifies both value-added and non-value-added time. For this project, the main source of variability revolves around the customer arrival patterns.

This value fluctuates due to the nature of Black Friday, with customers eager to have first access to the sale. Opening hours are typically more congested than later in the day, and by arriving earlier, customers are more likely to find stock available and minimize wait times. In this retail service system, Work in Process (WIP) refers to the customers currently moving through the process, from the time they enter the store until they complete checkout. Similar to "unfinished products" in manufacturing, customers are the units in production in a service setting. The largest WIP accumulation occurs during peak hours when many customers are simultaneously browsing and queuing, leading to longer flow times and increased congestion in high-traffic departments. By applying Lean concepts, particularly those focused on identifying waste and improving flow, this VSM highlights bottlenecks such as queue buildup at checkout due to high arrival variability and department-specific cashier constraints.

## Proposed Solutions

To reduce queue times and total flow time per customer on this busy holiday, the team wants to add additional capacity to the service stations. While it may be difficult to hire more physical workers, the team suggests first re-allocating current workers to departments needing the most assistance, or by installing self-checkout machines near current servers. This would allow one worker to oversee multiple checkout processes at one time, increasing the number of customers who can be served at one time. Based on current clusters of servers, we would place self-checkout machines in the North area of the second floor (near men's department), also at the Northeast and Southeast corners of the first floor. However, the overall goal for the moment is to simply increase capacity, in whichever way is most efficient, to meet the needs of customers entering the store.

To help manage the intensity of Black Friday crowding, the team proposes introducing a constant arrival rate system that reshapes how customers enter the store rather than simply reacting to congestion after it forms. Instead of allowing the traditional "opening-rush," where hundreds of shoppers enter simultaneously and immediately overwhelm specific departments, the constant arrival rate approach focuses on pacing the initial entry flow to create a more stable operating environment for employees and customers. By releasing customers in steady, predictable waves, the store can prevent the abrupt demand spikes that often cause long travel paths, excessive queue buildup, and reduced shopping efficiency. This method also gives staff time to reset racks, maintain organization, and provide better customer assistance, since they are no longer forced into crisis mode during the first hour of operation. Additionally, a controlled entry pattern helps balance demand across the entire store layout; customers spread out more naturally, departments are less likely to hit capacity all at once, and servers can maintain more consistent productivity. The goal is not just to slow arrivals, but to transform the start-of-day traffic into a smoother, more manageable flow that supports better decision-making, improved service quality, and an overall safer and more efficient shopping experience for everyone.

# Simulation Model

## Base Model

Macy's is part of the Mall of Louisiana's 1.5 million square feet of shopping space. The Macy's is 16,000 square feet, with seven different departments and hundreds of items for sale – everything from household goods to ties! The issue at hand is the customer processing system during the Black Friday holiday. After speaking with management, the team was told that the store operates from 6AM to 11PM on Black Friday and sees annually 35,000 guests. The goal of this project is to optimize customer shopping and checkouts by standardizing the Black Friday process. The base model setup and results are noted below, and inefficiencies have been highlighted to demonstrate the need for alternative solutions.

The base model contains seven departments, with 26 service stations. During the holiday season, seasonal workers are hired, in addition to standard staff, to provide extra support at checkout stations. The servers indicate cashier stations, and the capacity indicates the number of workers at each station.

- Men's and children's clothing: 4 servers (capacity 2 each)
- Furniture/household appliances: 2 servers (capacity 4 each)
- Macy's Backstage: 4 servers (capacity 2 each)
- Women's clothing: 3 servers (capacity 2 each)
- Shoes and junior's clothing: 2 servers (capacity 4 each)
- Jewelry: 5 servers (capacity 2 each)
- Perfume: 6 servers (capacity 2 each)

There are four model customers, two men and two women, who populate the system. Each entity has the same interarrival rate, based on numbers provided by Macy's management. Management stated that the hours of 8:00AM - 9:00AM and post-lunch hours (12:00PM - 2:00PM) see the most customer arrivals. While specific interarrival times were not known, the team estimated arrival rates per hour and then applied a sensitivity analysis to test different variances of arrival totals.

The hours with the highest arrival rates were highlighted, and every hour was assigned a percentage of total arrivals. From here, the total arrivals per hour to reach 35,000 customers were

calculated. Each value was then divided by four and inputted into a rate table within Simio, to be referenced by each entity. After running the model with 35,000 people, almost half of all entities were not leaving the system. The team decided that 35,000 was a high estimate for the total customers purchasing items on Black Friday within this specific Macy's, due to several factors such as groups of people checking out as one unit, some people only window shopping, and even management inflating numbers. Therefore, the team then calculated 20%, 30%, and 40% reductions of all original numbers. After testing arrival totals of 28,000; 24,500; and 21,000; the 40% reduction produced the most realistic results. While 35,000 can be a good estimate of the number of visitors in the system, not all will be purchasing items. Many may be children, grandparents, or friends who may be tagging along with the shopper. The value of 21,000 customers was an accurate starting point for analysis of the system.

*Table 2: Interarrival Times*

| Hours | Total entering % | Entities per arrival (across all 4 entities) | 20% Sensitivity Analysis | 30% Sensitivity Analysis | 40% Sensitivity Analysis | Entities per Arrival per Entity |
|---|---|---|---|---|---|---|
| 6AM-7AM | 0.05 | 1750 | 1400 | 1225 | 1050 | 262.5 |
| 7AM-8AM | 0.05 | 1750 | 1400 | 1225 | 1050 | 262.5 |
| 8AM-9AM | 0.2 | 7000 | 5600 | 4900 | 4200 | 1050 |
| 9AM-10AM | 0.08 | 2800 | 2240 | 1960 | 1680 | 420 |
| 10AM-11AM | 0.06 | 2100 | 1680 | 1470 | 1260 | 315 |
| 11AM-12PM | 0.06 | 2100 | 1680 | 1470 | 1260 | 315 |
| 12PM-1PM | 0.09 | 3150 | 2520 | 2205 | 1890 | 472.5 |
| 1PM-2PM | 0.08 | 2800 | 2240 | 1960 | 1680 | 420 |
| 2PM-3PM | 0.07 | 2450 | 1960 | 1715 | 1470 | 367.5 |
| 3PM-4PM | 0.05 | 1750 | 1400 | 1225 | 1050 | 262.5 |
| 4PM-5PM | 0.05 | 1750 | 1400 | 1225 | 1050 | 262.5 |
| 5PM-6PM | 0.05 | 1750 | 1400 | 1225 | 1050 | 262.5 |
| 6PM-7PM | 0.05 | 1750 | 1400 | 1225 | 1050 | 262.5 |
| 7PM-8PM | 0.04 | 1400 | 1120 | 980 | 840 | 210 |
| 8PM-9PM | 0.01 | 350 | 280 | 245 | 210 | 52.5 |
| 9PM-10PM | 0.01 | 350 | 280 | 245 | 210 | 52.5 |
| 10PM-11PM | 0 | 0 | 0 | 0 | 0 | 0 |
| Total Customers | 1 | 35000 | 28000 | 24500 | 21000 | 5250 |

All interarrival times highlighted in the rightmost column in Table 2 were placed into a rate table. The Data tab allows the user to select Rate Tables and create an hourly total of individuals to place in the system. All hourly arrivals were divided by four before being inputted into the table, to account for the four model entities within the process (Appendix Figure 3).

To build the model, the team went to Macy's and created a visual layout of the store. These images were then pulled into Simio and scaled to match the 16,000 square foot layout in meters (~40 meter per side) and locked in place (Appendix Figure 4).

Each model entity was given a specific sequence, to ensure men and women both had the opportunity to shop different departments. Appendix Figure 5 shows the sequence that each entity takes. Due to the endless possibilities of combinations for customer types (couples, families) it would be difficult to model every single possibility. The team has opted to model two

variants of a typical male or female shopper. Each entity has a specialized path that allows it to enter each department and check out with each before leaving the system. The sequence was selected to model customer behaviors and needs.

The model with one-directional paths is shown in Appendix Figure 6. All references within the sequence tables lead customers to shopping areas within each department. After shopping within an area, the entity is processed and goes to the next shopping area. To utilize the escalator, both images are connected by a connector with no time restraints, meaning that entities efficiently arrive at the next floor.

There is one sink at the base of the first floor, and all entities are exited out of this area. After running the model, it was found that the major issue was customer flowtimes, as well as getting all customers out of the system. The store struggles with checking out shoppers after closing. By simply asking these customers to leave, the franchise loses profit. However, to ensure all entities leave the system, the system must run for a total of 40 hours, which is 23 hours after the store's closing. While the real store would not stay open for that long to process customers, they do struggle with the number of customers remaining after closing hours. Instead, the real-world model will stay open an hour past closed, and then all customers will be asked to leave. For the case of this model, some statistics were pulled from the base model with its 40-hour flowtime.

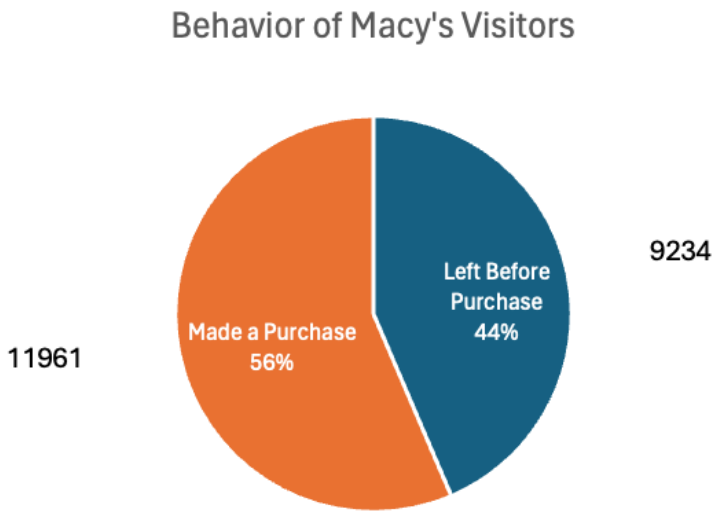



*Figure 4: Pie Chart of Customer Behavior*

A total of 21,195 people entered and left the system. 9,234 customers balked or reneged from the system without purchasing, during open store hours (Figure 4). The appendix shows

pivot results of the number of entities who entered and exited the store, per customer (Appendix Figure 10). Overall, customers spend an extended period of time in the Macy's as shown by output analysis of the system. Customers can spend, on average, 3 to 7 hours shopping and waiting in checkout lines.

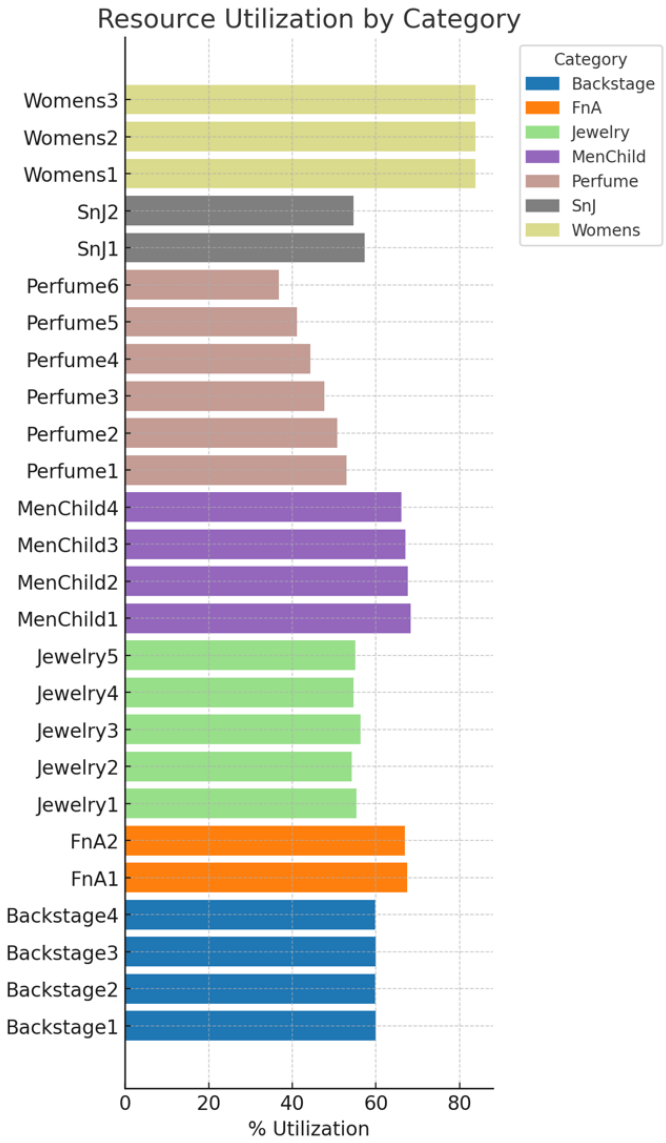



*Figure 5: Service Station Utilization by Department*

Scheduled utilizations for each server ranged between 36% - 80%, which is a high level of variance between departments (Figure 5) (Appendix Figure 11). Based on current conditions, not all workers are being utilized properly, mostly due to extended store hours to assist all customers with exiting the system.

The Macy's system could benefit from alternative adjustments that lower flowtimes for customers, allow for smoother shopping, and utilize employee resources accurately. Various improvement strategies are detailed as alternatives below.

## Description of Alternatives

A common alternative for the holiday would be to hire additional employees. Seasonal employees would be brought into Macy's for Black Friday and be trained for the influx of customers. Hiring managers would have to thoroughly review resumes and experiences for potential employees, ensuring that they are able to handle the input of customers and fast-paced environment. Since one of the eight wastes in lean is underused talent, the corporation would benefit from allocating skilled workers to their busiest departments; ideally, these workers would have shorter processing times. After analyzing the two entities that had the highest maximum flow time within the 40-hour period, (Man 2 with ~21 hours and Woman 2 with ~15 hours), the team analyzed what departments these two entities had in common. The team also analyzed the holding time for the paths that were extremely large, due to the current capacity not being enough and adjusted the server capacities that those paths were connected to. Each Women's, Backstage, Men and Children servers increased to a capacity of four, and Furniture increased to a capacity of six at each server.

Man 1 showed decreases of 59.8% and 65.4%; Man 2 showed decreases of 82.8% and 83.2%; Woman 1 showed decreases of 83.3% and 78.6%; and Woman 2 showed decreases of 71.9% and 66.5% in average and maximum flow times, respectively. The additional capacity proved to reduce maximum and average flowtimes to reasonable amounts, when compared with the base model (Appendix Figure 8).

Another alternative the team explored was adjusting the system to use a constant arrival rate for all customers. Using the estimate of 21,000 total shoppers, the team first divided this volume evenly among the four customer types, resulting in 5,250 customers per type. Because of the simulation models' 17 hours of store operation, this value was further divided across the full operating period, creating a constant arrival rate of approximately 309 customers per hour for each customer type. In real life, this would be monitored by police outside of the department store, only allowing 309 customers in per hour. Theoretically, majority of the 309 customers would be able to leave before the next batch of customers came in. Using the base model

capacities and changing the rate table to populate 309 entities per hour until the store closes, the team concluded that flowtimes decreased. However, the change is not significant enough, as Man 2 and Woman 2 still have an irregular maximum time in system, indicating that this is not the most effective alternative (Appendix Figure 9).

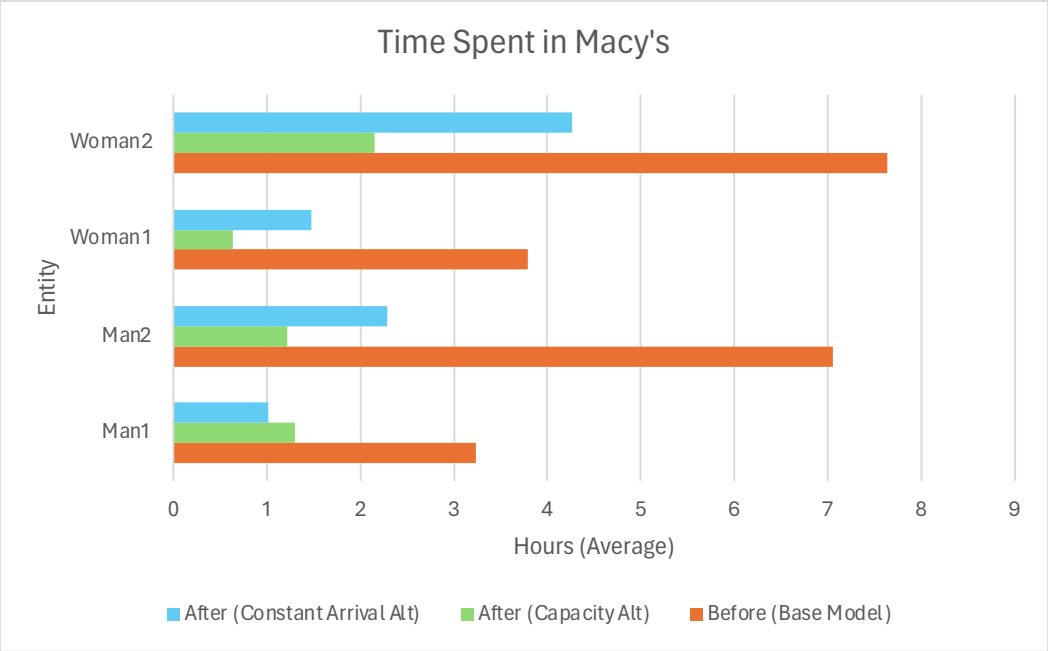



*Figure 6: Time Spent in System by Customer Type and Model Improvement*

All flow times per customer are visualized in Figure 6. The alternative producing the lowest average flow times is the capacity alternative. Statistical analysis was performed to prove the significance of flow time decrease of the capacity alternative, detailed in the Output section.

## Process Logic

Source W1, W2, M1, M2 all have their respective entities in Entity Type (Woman 1, Woman 2, Man 1, Man 2), Arrival Mode as Time Varying Arrival Rate, and the Rate Table is Rate Table 1. The team created a Probabilistic Balk Decision Type, with a probability of 0.35 or 35% of customers walking in and walking out (the store is too crowded or unable to find anything to purchase). The Balk Node Name is Input@Sink1.

After entering the store, each entity travels to a Basic Node, which would represent the entity shopping. Each node within the sequence has an Add – On delay process, which means that as customers enter the node, they sit for a specified amount of "shopping time." Shopping estimates were created based on trends from observing shoppers at Macy's. Due to the fast-paced

nature of Black Friday, customers do not spend as long in the store as they may on a normal day. Instead, customers know what deals are available and where they need to shop to find exactly what they are looking for. The delay times for each node are listed in the Appendix (Equation List 1).

After the entity shops, it goes to a transfer node via path that has the following process logic. Each transfer node has Entity Destination Type: Select From List, and the Node List Name is the respective node list. For each department, there is a node list that lists all of the input nodes of possible servers. For example, Shoes and Juniors (SnJ) has two servers, so the node list is Input@SnJ1 and Input@SnJ2.) The Selection Goal is Smallest Value, and the Value Expression is Candidate.Node.AssociatedStationLoad. This ensures that customers are able to pick the shortest line to queue.

At each server, there is a Processing Time of Random.Triangular (1,2,3) minutes. The input buffer capacity is 50. For Balking & Reneging Options, the Balk Decision Type is Blocked, and the Balk Node Name is Input@Sink1. For Renege Triggers, the Trigger Type is Time Based, and the Wait Duration is 1 hour, indicating that people will leave the line if they have been waiting for more than an hour. The Renege Node Name is Input@Sink1. The output of each server then leads to another basic node, which then is connected to a path that leads them to the next destination, following the respective sequence table. At the end of each sequence, the Input@Sink1 ensures the entities exit the department store.

## Validation and Verification

Due to the nature of this project and the inability of the team to collect exact data from an event that has not occurred for this year, different measures were implemented to create a realistic model. Data for Macy's Black Friday event specifically is also not published to the public, so the team can only use information from management combined with assumptions and estimates from personal anecdotes.

The model was validated by different methods, all of which proved that the model was acting similar to that of the real world, by examining the number of entities leaving the system between the base model and the alternative created. Validation occurs when the user is ensuring that the model applies to the real-world system being modeled. The focus for this project's

validation was the total numbers of entities within the system, as well as flowtime and throughput of each customer.

Within the original model, not all entities were leaving the store at the end of the day. Instead, the model needed to run additional hours to ensure all entities were serviced. In reality, not all customers are serviced on busy days, such as Black Friday. The model was showing a legitimate error that occurs during Black Friday sales at the Baton Rouge Macy's. This output demonstrated consistency between the base model and the real-life situation. The output also passes the validation test of symptom generation, which means that the simulation model accurately shows difficulties that occur in the real system.

Consistency was confirmed when the model was ran multiple times and the same value was obtained for the number of entities created and left to the sink. Degeneracy was also investigated by slowly changing the capacities of the servers. With a small increase in capacity for each server, there was a reduced flowtime for the entities and a lower scheduled utilization for the respective servers. There are no absurd conditions, such as zero flow times or infinite values or queues and flow times.

One of the last aspects of validation was statistical analysis of the output values. After running the base model, the following confidence intervals were found for each entities' flowtime (Table 3).

*Table 3: Confidence Interval by Customer Type for Base Model (Flowtime)*

| Entity | Average | Half Width | CI Lower Bound | CI Upper Bound |
|---|---|---|---|---|
| Man 1 | 3.3949 | 0.1133 | 3.2816 | 3.5082 |
| Man 2 | 6.8577 | 0.1 | 6.7577 | 6.9577 |
| Woman 1 | 3.6584 | 0.0424 | 3.616 | 3.7008 |
| Woman 2 | 7.77308 | 0.15811 | 7.61497 | 7.93119 |

The Half Width serves as the plus or minus value to the mean flowtime, showing the range of flowtimes that match predicted values, up to 95%. The flowtimes and half widths were calculated by running a Simio experiment with 10 replications of a 17-hour day. All half widths are within 2.544 minutes (0.0424 hours) to 9.486 minutes (0.1581 hours) of the mean flowtime value. With such large flowtimes in the base model, which show the processing error within the system, we can ensure with 95% confidence the full extent of variability within entity shopping time in the system. These outputs match verbal data collected from Macy's management. Management said that lines have high queues and there are many customers who shop for

extended periods of time on this day. While exact numbers were not recorded or known, the confidence intervals helps the team ensure that all possible values of flowtimes are captured within the base model.

The alternative test was also run as an experiment with 10 replications, so that corrected flowtimes and their confidence intervals could be obtained (Table 4).

*Table 4: Capacity Alternative Confidence Intervals (Flowtime)*

| Entity | Average | Half Width | CI Lower Bound | CI Upper Bound |
|---|---|---|---|---|
| Man 1 | 1.2127 | 0.0431 | 1.1696 | 1.2558 |
| Man 2 | 1.2059 | 0.0238 | 1.1821 | 1.2297 |
| Woman 1 | 0.6406 | 0.0079 | 0.6327 | 0.6485 |
| Woman 2 | 2.0987 | 0.0495 | 2.0492 | 2.1482 |

By implementing the alternative of adjusting capacities and reducing variability in arrival time, the team was able to reduce flowtimes significantly, while also shrinking the confidence interval range. The smaller range reduces variability and ensures a more accurate prediction of flow times within the system.

The last aspect of validation is ensuring the structure of the model reflects the real-world model. The base model is based realistically on the behavior of Macy's on Black Friday according to observations and management input. It consists of 26 servers with a capacity of 2 to 4 to represent the regular retail workers as well as temporary ones hired for the holiday season. The processing times reflect the ones provided to us by management. A rate table is used to determine the interarrival times of each hour for each entity type. The interarrival times are also based on information provided to us by management. Sequence tables are used to reflect each entity type's path which was determined by observations in the store. Paths flow between the sequence orders to reflect the orders of each entity type. Paths connect to a BasicNode which represents the entities shopping. They are modeled as delays in processes. Paths branch off from these nodes to another BasicNode that allows the entity to choose the shortest line to the associated checkout servers. The lines converge to another BasicNode that acts as the exit from that department to the next department in the sequence. All entities eventually flow into the sink, which represents exiting the store.

The model was also verified, meaning that the logic of the model is accurate, by multiple tests. One specific strategy was the use of Model Trace on Simio. This feature runs continuously, recording every event so the user can easily see any errors and perform root cause analysis on the

system. This makes it possible to see when or if an individual entity entered a certain input node, was created or destroyed, when it transfers from a certain server to an output node, and so on. The base model did not have any errors detected by the trace, ensuring that there were no logic errors with the system. A sample of the trace can be found in Appendix Figure 12.

Performing a systematic walk through of the model's logic to confirm the process matched the conceptual design was done by getting rid of all the entities besides one and watching the routing logic it was performing. Running the model in simplified conditions tests whether outputs and entities behave predictably. Visually, the entity was moving through the correct nodes in the correct order, there were no entities teleporting or walking in free space. The paths were also checked to see if they were in the correct direction for the entity's movement. While doing this there were no errors or warnings that popped up to inform the team that something was invalid or missing. When building the model, we incrementally built out the sequence paths of each entity to make sure paths between nodes were attached to the correct destinations. The team checked that the correct add on processes were assigned to the appropriate nodes they were designed for. Checks for unit consistency in the processing and interarrival times took place, as well as numbers in the rate table.

Overall, the model is both verified and validated to match the real-world Macy's Black Friday issue, so that the company can see and possibly implement alternatives that utilize lean methodology to better the system.

# Output Analysis

## Experiments

Similarly to the previous restaurant lab problem conducted, no warm up period was used, since statistics would be needed from the very beginning of opening the store up until now. 10 replications were run at a run length of 17 hours. Macy's is open from 6:00 AM to 10:00 PM on Black Friday, which is 16 hours, but an extra hour is added so that the remaining entities can finish their transactions and exit the system. This is a common practice for many stores; if customers are in line before closing, they will be served.

The team added capacity controls for each server using real properties (Appendix Figures 13 and 14). This allows multiple scenarios to be run with varying capacities for each server. The

properties were named after the server, for example: BackstageCap, JewelryCap, WomensCap, etc. Since the capacities amongst the servers in each departments were kept consistent, only one property was created for each department.

Due to the number of servers in the system, it would be inefficient to add all of the servers as responses. The team would have to scroll left and right, and it would not be as effective as analyzing from the pivot grid after filtering. However, if the team were to add responses for everything, the following expressions would be used. Backstage 1 and Woman 1 are used as an example, as shown in the appendix, but each server and entity would have its respective name.

The specific data used for statistical analysis was cycle times, which were set as responses for each scenario. The use of responses allows for instant data collection, based on experiments ran. Figures 7-9 show the specific responses added, to collect cycle times per customer, per model. These times were then used to determine statistical significance.

| Responses | | | |
|---|---|---|---|
| Woman1CT | Woman2CT | Man1CT | Man2CT |
| 3.65843 | 7.7308 | 3.39492 | 6.85769 |

*Figure 7: Flow Time Responses for Base Model*

| Responses | | | |
|---|---|---|---|
| Woman1CT | Woman2CT | Man1CT | Man2CT |
| 0.640557 | 2.09913 | 1.2127 | 1.20587 |

*Figure 8: Flow Time Responses for Capacity Alternative*

| Responses | | | |
|---|---|---|---|
| Woman1CT | Woman2CT | Man1CT | Man2CT |
| 1.19336 | 2.52155 | 1.0117 | 1.22929 |

*Figure 9: Flow Time Capacity for Constant Arrival Alternative*

## Statistical Results

Using pivot results from each experiment within the base model, the capacity alternative, and the constant arrival alternative, assumptions can be drawn as to which alternative is the best option to reach operation goals. The performance measures that have been observed within this study have been server utilization, and most importantly, customer flow times. Below, server utilizations for each model are shown, as well as their corresponding half widths. The half width shows the plus and minus value to determine 95% confidence intervals. For example, the Backstage 1 server has an average server utilization of 58%, and the total confidence interval would be the average value, plus/minus the half width of 0.2477%. This would lead to a confidence interval of 58.51% -59.01% for the first Backstage server.

*Table 5: Utilization for Base, Capacity, Const. Arrival Models*

| Server | Base Model Avg Util | Base Model Half Width | Cap Alt Avg Util. | Cap Alt. Half Width | Const. Arrival Alt Avg Util. | Const. Arrival Alt Avg Util. |
|---|---|---|---|---|---|---|
| Backstage 1 | 58.7631 | 0.2477 | 80.509 | 0.4438 | 46.2023 | 0.3981 |
| Backstage 2 | 58.6985 | 0.2688 | 78.7157 | 0.4708 | 46.1635 | 0.4003 |
| Backstage 3 | 58.7008 | 0.2704 | 76.9425 | 0.6242 | 46.0959 | 0.388 |
| Backstage 4 | 58.6654 | 0.2866 | 75.3577 | 0.6961 | 46.0682 | 0.4014 |
| Jewelry 1 | 55.7167 | 0.8261 | 80.6585 | 1.2704 | 46.0488 | 0.4194 |
| Jewelry 2 | 55.4568 | 0.6733 | 81.208 | 1.2272 | 46.2544 | 0.3763 |
| Jewelry 3 | 55.5732 | 0.9047 | 81.153 | 0.9062 | 46.0923 | 0.2917 |
| Jewelry 4 | 55.1532 | 0.3176 | 80.8443 | 0.7818 | 46.1842 | 0.3866 |
| Jewerly 5 | 54.879 | 0.9659 | 81.4866 | 1.0935 | 46.1669 | 0.3129 |
| Womens 1 | 82.2256 | 0.4421 | 87.5064 | 0.5279 | 64.6763 | 0.7817 |
| Womens 2 | 82.2334 | 0.4719 | 86.7298 | 0.5101 | 64.6184 | 0.7818 |
| Womens 3 | 82.1681 | 0.4593 | 85.9099 | 0.5352 | 64.6305 | 0.7966 |
| SnJ 1 | 56.6895 | 0.3988 | 87.3494 | 0.4841 | 64.6501 | 0.7648 |
| SnJ 2 | 54.2301 | 0.4121 | 86.4554 | 0.492 | 64.611 | 0.7792 |
| Perfume 1 | 53.0281 | 0.3174 | 76.6054 | 0.419 | 43.4664 | 0.2998 |
| Perfume 2 | 50.7842 | 0.342 | 73.5078 | 0.5408 | 41.8918 | 0.2168 |
| Perfume 3 | 48.206 | 0.3275 | 69.7247 | 0.3787 | 40.0279 | 0.2518 |
| Perfume 4 | 45.1416 | 0.3513 | 65.9349 | 0.3869 | 37.7652 | 0.2911 |
| Perfume 5 | 41.7023 | 0.2386 | 61.8388 | 0.3549 | 35.3278 | 0.2015 |
| Perfume 6 | 38.0563 | 0.241 | 57.5726 | 0.4713 | 32.2969 | 0.2197 |
| MenChild 1 | 68.5939 | 0.4893 | 84.4839 | 0.661 | 56.7673 | 0.5266 |
| MenChild 2 | 68.0253 | 0.5108 | 82.7404 | 0.6337 | 54.4969 | 0.5206 |
| MenChild 3 | 67.4736 | 0.5486 | 80.6846 | 0.7401 | 52.5383 | 0.5229 |
| MenChild4 | 66.9187 | 0.6164 | 78.3969 | 0.7843 | 50.4581 | 0.5625 |
| FnA1 | 67.8719 | 0.5525 | 89.3505 | 0.6593 | 54.4332 | 0.5269 |
| FnA2 | 67.5681 | 0.611 | 85.5615 | 0.8512 | 52.7289 | 0.5427 |

All average utilizations have been visually sorted through Excel color scales, with the highest values being a darker, red color (Table 5). The initial base model had low server utilizations, which was mostly caused by the extended run time of the system. Not all customers would exit the system by the end of the store day, mostly due to long wait times and customer flow times. The capacity alternative aimed to shift resources, while also adding additional capacity, to minimize flow times. The total employees changed from 18 to 35, with adding

servers to places such as Furniture and Appliances and Shoes and Juniors seeing the highest increase in capacity. Shifting focus to the average server utilizations for the Capacity Alternative: percent utilization increased to higher levels per server. These levels are still acceptable for workers, none of which exceed 90% on average, and the server capacity allows workers to fully check out all customers within the 17-hour shift. The second alternative, creating constant arrival rates, had significantly lower utilizations. Creating a system where arrivals lack variability is beneficial for workflow, but by using the same service capacity, the team does not have the resources to serve all customers entering the system.

The flow times, also denoted as cycle times, were collected for each entity in every model. The half widths were also collected to determine the 95% confidence interval of all values. The model showing the best capability to better the Macy's system is Alternative 1 (Second Table in Table 6). The increase in server capacity leads to immensely shorter flow times. Workers are more efficient with this extra assistance, which increases sales and overall customer turnover.

*Table 6: Confidence Intervals and Averages for Cycle Time*

| Entity | Base CT (Hr) | Base Half Width | CI Lower Bound | CI Upper Bound |
|---|---|---|---|---|
| Woman 1 | 3.65843 | 0.0424 | 3.61603 | 3.70083 |
| Woman 2 | 7.7308 | 0.1581 | 7.5727 | 7.8889 |
| Man 1 | 3.39492 | 0.1133 | 3.28162 | 3.50822 |
| Man 2 | 6.85769 | 0.1 | 6.75769 | 6.95769 |
| | | | | |
| Entity | Cap Alt CT (Hr) | Cap Alt Half Width | CI Lower Bound | CI Upper Bound |
| Woman 1 | 0.847293 | 0.0322 | 0.815093 | 0.879493 |
| Woman 2 | 1.87456 | 0.0423 | 1.83226 | 1.91686 |
| Man 1 | 0.970603 | 0.036 | 0.934603 | 1.006603 |
| Man 2 | 1.48735 | 0.0334 | 1.45395 | 1.52075 |
| | | | | |
| Entity | Const. Arriv Alt CT (Hr) | Const. Arriv Alt Half Width | CI Lower Bound | CI Upper Bound |
| Woman 1 | 1.34642 | 0.0624 | 1.28402 | 1.40882 |
| Woman 2 | 4.13593 | 0.1179 | 4.01803 | 4.25383 |
| Man 1 | 1.0711 | 0.1045 | 0.9666 | 1.1756 |
| Man 2 | 2.21583 | 0.0496 | 2.16623 | 2.26543 |

The smaller the half width, the smaller the confidence interval. In the case of Alternative 1, all half widths are within the hundredths place in value, meaning the range of possible values is tight. Flowtimes for this alternative are likely to be within this small range, which leads to more accurate estimation of performance indicators. Further statistical analysis can be completed through specific tests for significance, noted below.

## Alternatives Statistics

After creating all alternatives, statistical analysis was done to numerically compare alternatives. The main performance indicator of this study is the average flow time of the customer. By minimizing flow time, customers spend less time in the store. As stated previously, Macy's struggles with having all customers check out by the time store closes. If they are unable to make all customers leave, then workers either stay longer, or customers are forced to leave without making purchases, which affects profit. Also, by reducing flow time, the customers will have shorter waits in lines and more satisfaction with their shopping experience. An experiment for the base model, the increased capacity alternative, and the constant arrival rate alternative were run with 10 replications. Run lengths were based on the minimum number of hours it would take to have all customers exit the system. The only model to successfully release all customers at the end of the 17-hour workday was the capacity alternative. However, all alternative flow times per each customer type were compared to the base model, using a Paired T-Test.

To obtain data, response results were exported from Simio for each model, which included cycle times per replication for Woman 1, Woman 2, Man 1, and Man 2. Each customer type was analyzed separately, and the difference between replications of the base model and alternative 1 (Capacity) and the base model and alternative 2 (Constant Arrivals) were calculated using Excel.

*Table 7: Paired T-Test on Replications for W1 (Flowtime)*

| Replication | Base W1 | Cap. Alt W1 | Const. Alt W1 | Difference of Cap Alt and Base | Difference of Const Alt and Base |
|---|---|---|---|---|---|
| 1 | 3.609815703 | 0.878973122 | 1.4726922 | -2.730842581 | -2.137123503 |
| 2 | 3.596140817 | 0.883543602 | 1.351306305 | -2.712597215 | -2.244834512 |
| 3 | 3.609019637 | 0.934024199 | 1.383862403 | -2.674995437 | -2.225157234 |
| 4 | 3.661448989 | 0.871034789 | 1.296576315 | -2.7904142 | -2.364872674 |
| 5 | 3.608000106 | 0.812766334 | 1.42471252 | -2.795233772 | -2.183287586 |
| 6 | 3.681104824 | 0.798888349 | 1.212272115 | -2.882216475 | -2.468832709 |
| 7 | 3.725908765 | 0.800862634 | 1.391887793 | -2.925046131 | -2.334020972 |
| 8 | 3.778384841 | 0.855658159 | 1.204647963 | -2.922726681 | -2.573736877 |
| 9 | 3.636508345 | 0.80450042 | 1.332605541 | -2.832007925 | -2.303902804 |
| 10 | 3.677989434 | 0.832679898 | 1.393680926 | -2.845309536 | -2.284308508 |
| **AVG** | | | | -2.811138995 | -2.312007738 |
| **StDev** | | | | 0.086575177 | 0.131831989 |

The next steps after finding mean difference values and standard deviations of the data set using Excel, was to find the standard error, which is the standard deviation divided by the square root of the number of replications (Table 7 and 8).

*Table 8: Standard Error and Confidence Intervals of Replications (Flowtime)*

| Standard Error (Standard Deviation / Sqrt(n)) | | | t value for 95% CI |
|---|---|---|---|
| Cap Alt | 0.027377 | | 2.262157 |
| Const Alt | 0.041689 | | |
| | | | |
| **Confidence Interval (Avg D +/- SE)** | | | |
| | Upper CI | Lower CI | |
| Cap Alt | -2.78376 | -2.83852 | |
| Const Alt | -2.27032 | -2.3537 | |

Using a t-value of 2.26 for 95% confidence, the upper and lower confidence intervals were calculated by taking the average distance, plus/minus the standard error. All confidence intervals fell into the negatives, meaning that the alternatives show significant improvement in lowering average flow times for each customer. All confidence intervals for alternative 1 were more negative than confidence intervals for alternative 2, signaling that alternative 1 is the best choice for decreasing flow times. All calculations for the remaining customers are listed in the appendix.

## Challenges

A major issue that the team ran into was the base model of the simulation. After adding in the values provided by management, the system would pop up several errors, a main one warning that there were more than 20,000 active entities in the system. With the original numbers, the number of entities exiting the system was not consistent with the number of entities we allocated in the rate table. The team attempted to debug the model by starting over, redoing each path to ensure that there were adequate connections for each sequence table, and redoing the delays in the add on processes. Since the model is handling an extreme number of entities, every small tweak would take a substantial amount of time to run and see the results of, even on a quicker speed setting and using the fast forward feature. After extensive trial and error, we were able to have all the entities leave the store, modeling real life circumstances.

Another issue would revolve around the numbers created for the simulation model. With a large-scale project modeling a department store, there is no exactness in the number for checkout/service (processing) times, shopping/browsing times (add on process delays), which

have an impact on nearly everything in the system. For example, a larger browsing time may be connected to a larger service time, or it could end up with the customer leaving due to a multitude of reasons: not finding anything, the lines are too long, etc. There are too many possibilities to account for, which is why this simulation model can only reasonably depict some situations that encompass common scenarios, and not every single possible scenario. Many customer sequences were unable to be accounted for directly. The thousands of shoppers can easily have different paths they take throughout the store, but this simply cannot be feasible given the team's time and resource constraints. However, by using generated data from the information provided, the team was able to analyze a large scale problem and apply engineering approaches.

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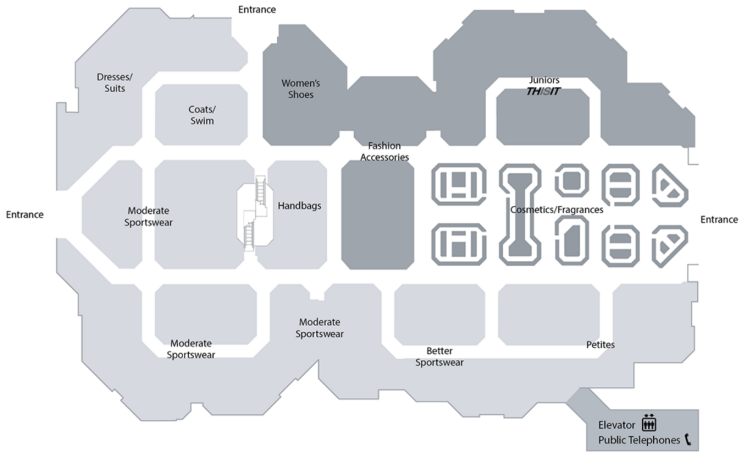
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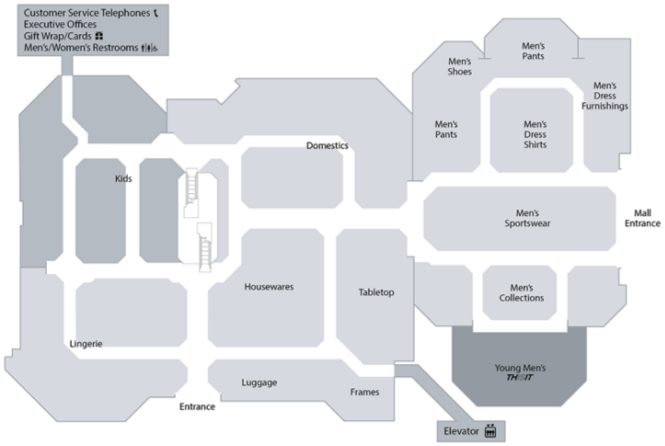
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# Appendix



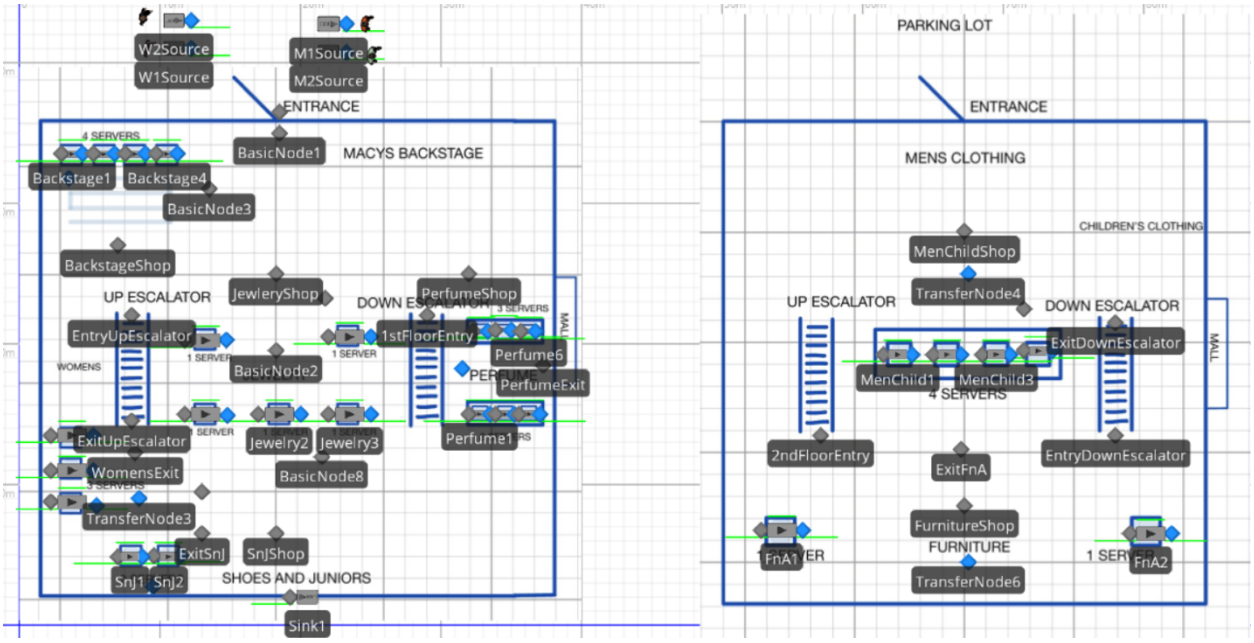

*Appendix Figure 1: First Floor Generic Macy's Model*




*Appendix Figure 2: Second Floor Generic Macy's Model*

| Starting Offset | Ending Offset | Rate (events per hour) |
|---|---|---|
| Day 1, 00:00:00 | Day 1, 01:00:00 | 262 |
| Day 1, 01:00:00 | Day 1, 02:00:00 | 262 |
| Day 1, 02:00:00 | Day 1, 03:00:00 | 1050 |
| Day 1, 03:00:00 | Day 1, 04:00:00 | 420 |
| Day 1, 04:00:00 | Day 1, 05:00:00 | 315 |
| Day 1, 05:00:00 | Day 1, 06:00:00 | 315 |
| Day 1, 06:00:00 | Day 1, 07:00:00 | 472 |
| Day 1, 07:00:00 | Day 1, 08:00:00 | 420 |
| Day 1, 08:00:00 | Day 1, 09:00:00 | 367 |
| Day 1, 09:00:00 | Day 1, 10:00:00 | 262 |
| Day 1, 10:00:00 | Day 1, 11:00:00 | 262 |
| Day 1, 11:00:00 | Day 1, 12:00:00 | 262 |
| Day 1, 12:00:00 | Day 1, 13:00:00 | 262 |
| Day 1, 13:00:00 | Day 1, 14:00:00 | 210 |
| Day 1, 14:00:00 | Day 1, 15:00:00 | 52 |
| Day 1, 15:00:00 | Day 1, 16:00:00 | 52 |
| Day 1, 16:00:00 | Day 1, 17:00:00 | 0 |

*Appendix Figure 3: Rate Table*




*Appendix Figure 4: Base Model Layout in Simio*

| | Sequence |
|---|---|
| 1 | BackstageShop |
| 2 | JewleryShop |
| 3 | PerfumeShop |
| 4 | Input@Sink1 |
| * | |

*W1*

| | Sequence |
|---|---|
| 1 | WomensShop |
| 2 | SnJShop |
| 3 | FurnitureShop |
| 4 | MenChildShop |
| 5 | Input@Sink1 |

*W2*

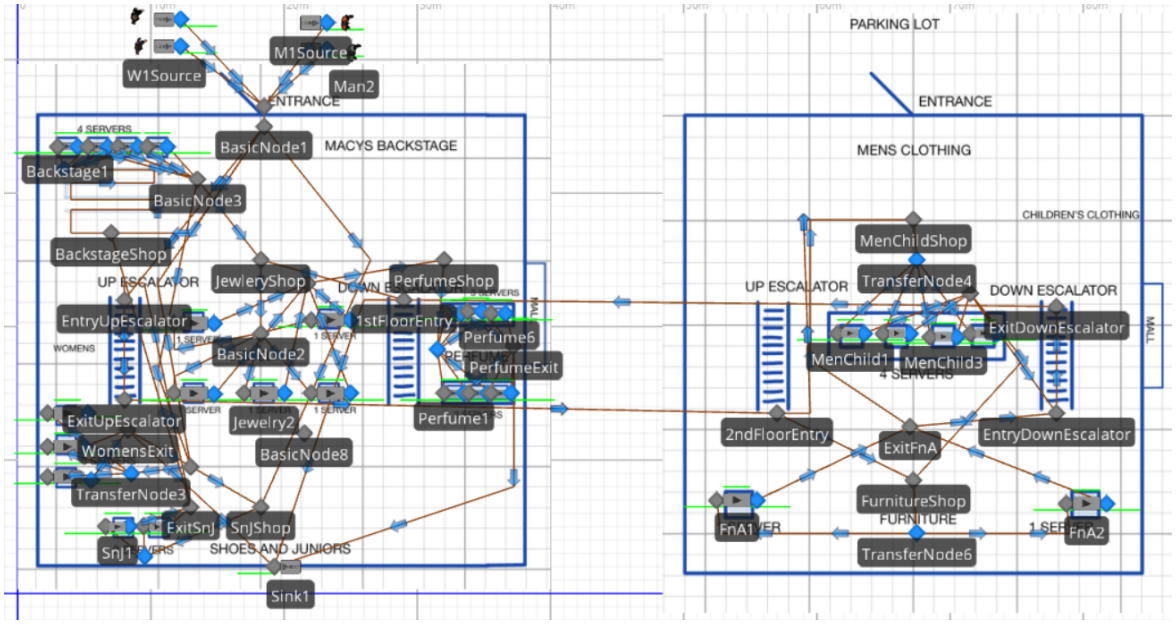
| | Sequence |
|---|---|
| 1 | MenChildShop |
| 2 | FurnitureShop |
| 3 | JewleryShop |
| 4 | PerfumeShop |
| 5 | Input@Sink1 |

*M1*

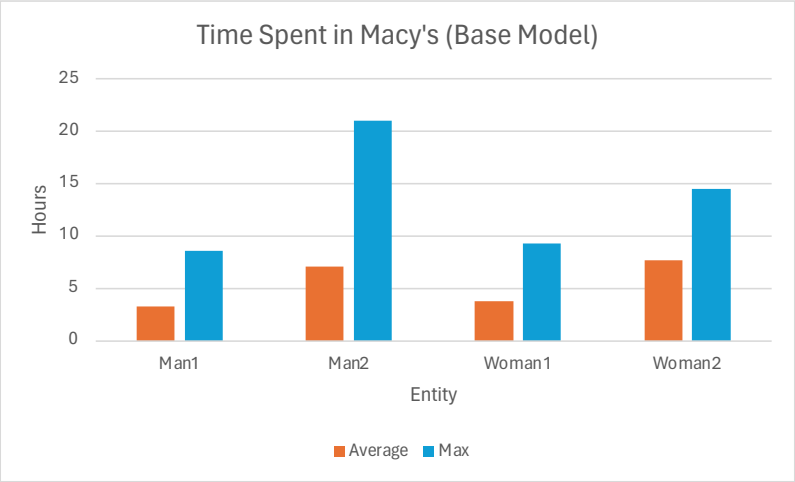
| | Sequence |
|---|---|
| 1 | SnJShop |
| 2 | BackstageShop |
| 3 | WomensShop |
| 4 | Input@Sink1 |

*M2*

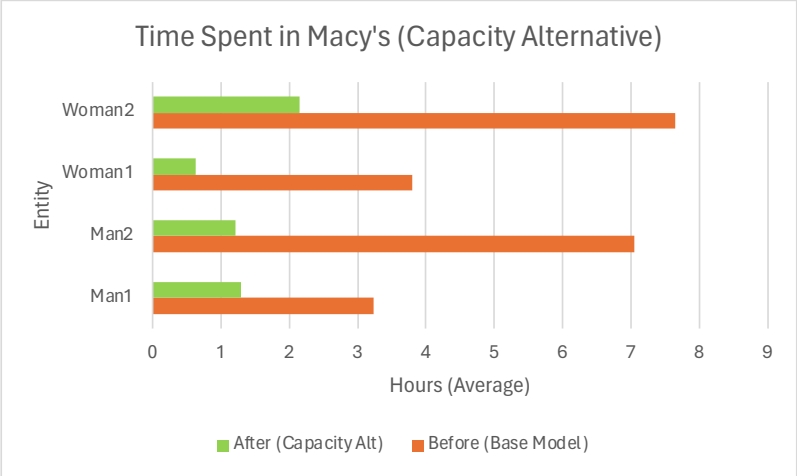
*Appendix Figure 5: Woman1, Woman2, Man1, Man2 Sequence Tables*



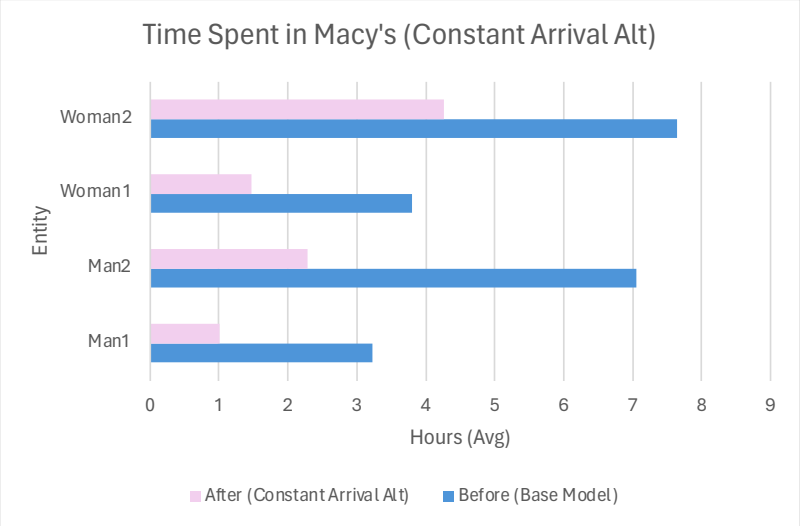

*Appendix Figure 6: Simio Base Model with Paths*




*Appendix Figure 7: Comparison of Avg and Max Time Spent within Macy's*




*Appendix Figure 8: Capacity Alternative Flowtime Result vs. Base Model Flowtime Results*




*Appendix Figure 9: Constant Arrival Alt. Flowtimes vs. Base Model Flowtimes*

Average

Drop Column Fields H

| Object Type | Object Name | Data Source | Category | Data Item | Statistic | Average Total |
|---|---|---|---|---|---|---|
| Source | M1Source | Processing | Throughput | NumberEntered | Total | 5,233.0000 |
| | | | | NumberExited | Total | 5,233.0000 |
| | M2Source | Processing | Throughput | NumberEntered | Total | 5,454.0000 |
| | | | | NumberExited | Total | 5,454.0000 |
| | W1Source | Processing | Throughput | NumberEntered | Total | 5,240.0000 |
| | | | | NumberExited | Total | 5,240.0000 |
| | W2Source | Processing | Throughput | NumberEntered | Total | 5,268.0000 |
| | | | | NumberExited | Total | 5,268.0000 |

| Object Type | Object Name | Data Source | Category | Data Item | Statistic | Average Total |
|---|---|---|---|---|---|---|
| Sink | Sink1 | InputBuffer | Throughput | NumberEntered | Total | 21,195.0000 |
| | | | | NumberExited | Total | 21,195.0000 |

*Appendix Figure 10: Simio Pivot Result of Total Customers Who Entered and Exited the Base Model*

| Object Type | Object Name | Data Source | Category | Data Item | Statistic | Average Total |
|---|---|---|---|---|---|---|
| Server | Jewelry1 | InputBuffer | Throughput | NumberBalked | Total | 55.0000 |
| | Jewelry2 | InputBuffer | Throughput | NumberBalked | Total | 57.0000 |
| | Jewelry3 | InputBuffer | Throughput | NumberBalked | Total | 36.0000 |
| | Jewelry5 | InputBuffer | Throughput | NumberBalked | Total | 95.0000 |
| | SnJ1 | InputBuffer | Throughput | NumberReneged | Total | 948.0000 |
| | SnJ2 | InputBuffer | Throughput | NumberReneged | Total | 943.0000 |
| Source | M1Source | OutputBuffer | Throughput | NumberBalked | Total | 1,820.0000 |
| | M2Source | OutputBuffer | Throughput | NumberBalked | Total | 1,912.0000 |
| | W1Source | OutputBuffer | Throughput | NumberBalked | Total | 1,810.0000 |
| | W2Source | OutputBuffer | Throughput | NumberBalked | Total | 1,558.0000 |

| Object Type | Object Name | Data Source | Category | Data Item | Statistic | Average Total |
|---|---|---|---|---|---|---|
| Server | Backstage1 | [Resource] | Capacity | ScheduledUtilization | Percent | 59.9312 |
| | Backstage2 | [Resource] | Capacity | ScheduledUtilization | Percent | 59.7944 |
| | Backstage3 | [Resource] | Capacity | ScheduledUtilization | Percent | 59.8759 |
| | Backstage4 | [Resource] | Capacity | ScheduledUtilization | Percent | 59.7402 |
| | FnA1 | [Resource] | Capacity | ScheduledUtilization | Percent | 67.4515 |
| | FnA2 | [Resource] | Capacity | ScheduledUtilization | Percent | 66.9116 |
| | Jewelry1 | [Resource] | Capacity | ScheduledUtilization | Percent | 55.3214 |
| | Jewelry2 | [Resource] | Capacity | ScheduledUtilization | Percent | 54.2349 |
| | Jewelry3 | [Resource] | Capacity | ScheduledUtilization | Percent | 56.2908 |
| | Jewelry4 | [Resource] | Capacity | ScheduledUtilization | Percent | 54.6860 |
| | Jewelry5 | [Resource] | Capacity | ScheduledUtilization | Percent | 55.0680 |
| | MenChild1 | [Resource] | Capacity | ScheduledUtilization | Percent | 68.3147 |
| | MenChild2 | [Resource] | Capacity | ScheduledUtilization | Percent | 67.6492 |
| | MenChild3 | [Resource] | Capacity | ScheduledUtilization | Percent | 67.0614 |
| | MenChild4 | [Resource] | Capacity | ScheduledUtilization | Percent | 66.0680 |
| | Perfume1 | [Resource] | Capacity | ScheduledUtilization | Percent | 52.9885 |
| | Perfume2 | [Resource] | Capacity | ScheduledUtilization | Percent | 50.7048 |
| | Perfume3 | [Resource] | Capacity | ScheduledUtilization | Percent | 47.6050 |
| | Perfume4 | [Resource] | Capacity | ScheduledUtilization | Percent | 44.3133 |
| | Perfume5 | [Resource] | Capacity | ScheduledUtilization | Percent | 41.1031 |
| | Perfume6 | [Resource] | Capacity | ScheduledUtilization | Percent | 36.7505 |
| | SnJ1 | [Resource] | Capacity | ScheduledUtilization | Percent | 57.3076 |
| | SnJ2 | [Resource] | Capacity | ScheduledUtilization | Percent | 54.7101 |

*Appendix Figure 11: Server Utilizations from Simio*

*Equation List 1*

*Basic Node 1 (**BackstageShop**)*

```
Math.If(
   ModelEntity.EntityType.Name == "Woman1", Random.Triangular(3, 5, 10),
 Random.Triangular(3, 4, 6))
```

*Basic Node 2 (**JewelryShop**)*

```
Math.If(
   ModelEntity.EntityType.Name == "Woman1", Random.Triangular(5, 10, 11),
 Random.Triangular(4, 7, 10))
```

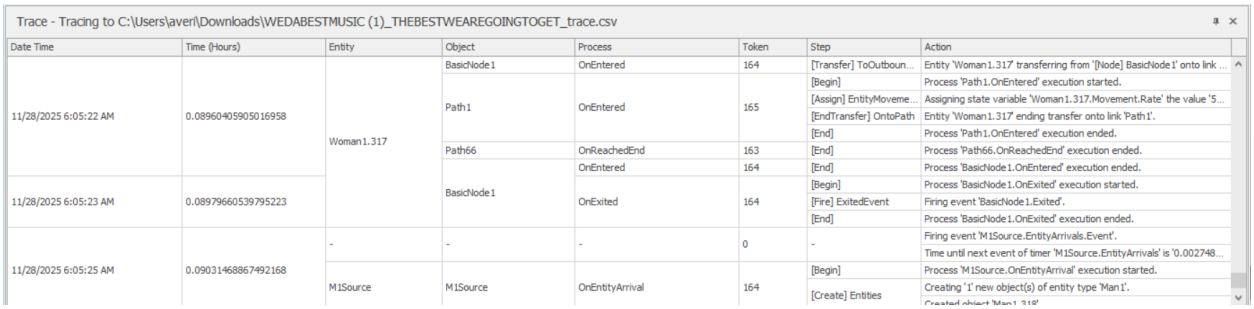
*Basic Node 3 (**WomensShop**)*
*Math.If(*
*ModelEntity.EntityType.Name == "Woman2", Random.Triangular(5, 7, 10),*
*Random.Triangular(3, 5, 6))*

*Basic Node 4 (**SnJShop**)*
*Math.If(*
*ModelEntity.EntityType.Name == "Woman2", Random.Triangular(3, 5, 10),*
*Random.Triangular(2,5,9))*

*Basic Node 5 (**PerfumeShop**)*
*Math.If(*
*ModelEntity.EntityType.Name == "Woman1", Random.Triangular(1, 3, 6),*
*Random.Triangular(1,2, 4))*

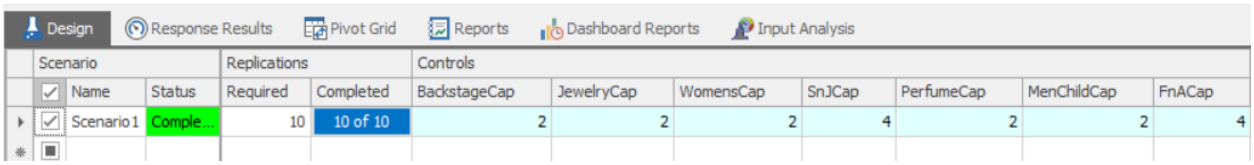
*Basic Node 6 (**Furniture Shop**)*
*Math.If(*
*ModelEntity.EntityType.Name == "Woman2", Random.Triangular(4, 5, 10),*
*Random.Triangular(5, 6, 7))*

*Basic Node 7 (**MenChildShop**)*
*Math.If(*
*ModelEntity.EntityType.Name == "Woman2", Random.Triangular(3, 5, 7),*
*Random.Triangular(3, 4, 6))*


Trace - Tracing to C:\Users\averi\Downloads\WEDABESTMUSIC (1)_THEBESTWEAREGOINGTOGET_trace.csv

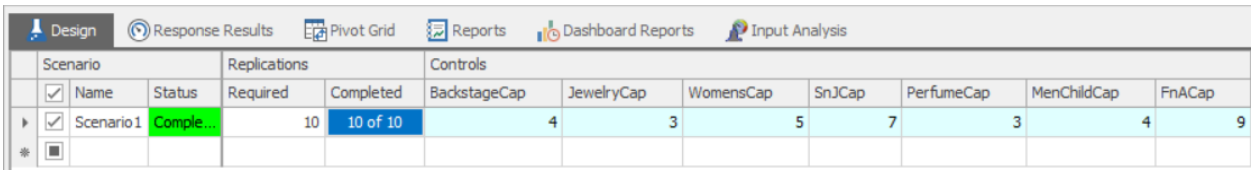
| Date Time | Time (Hours) | Entity | Object | Process | Token | Step | Action |
|---|---|---|---|---|---|---|---|
| 11/28/2025 6:05:22 AM | 0.08960405905016958 | Woman1.317 | BasicNode1 | OnEntered | 164 | [Transfer] ToOutboun... | Entity 'Woman1.317' transferring from '[Node] BasicNode1' onto link ... |
| | | | Path1 | OnEntered | 165 | [Begin] | Process 'Path1.OnEntered' execution started. |
| | | | | | | [Assign] EntityMoveme... | Assigning state variable 'Woman1.317.Movement.Rate' the value '5... |
| | | | | | | [EndTransfer] OntoPath | Entity 'Woman1.317' ending transfer onto link 'Path1'. |
| | | | | | | [End] | Process 'Path1.OnEntered' execution ended. |
| | | | Path66 | OnReachedEnd | 163 | [End] | Process 'Path66.OnReachedEnd' execution ended. |
| | | | BasicNode1 | OnEntered | 164 | [End] | Process 'BasicNode1.OnEntered' execution ended. |
| 11/28/2025 6:05:23 AM | 0.08979660539795223 | | | OnExited | 164 | [Begin] | Process 'BasicNode1.OnExited' execution started. |
| | | | | | | [Fire] ExitedEvent | Firing event 'BasicNode1.Exited'. |
| | | | | | | [End] | Process 'BasicNode1.OnExited' execution ended. |
| 11/28/2025 6:05:25 AM | 0.09031468867492168 | - | - | - | 0 | - | Firing event 'M1Source.EntityArrivals.Event'. |
| | | | | | | | Time until next event of timer 'M1Source.EntityArrivals' is '0.002748... |
| | | M1Source | M1Source | OnEntityArrival | 164 | [Begin] | Process 'M1Source.OnEntityArrival' execution started. |
| | | | | | | [Create] Entities | Creating '1' new object(s) of entity type 'Man1'. |

*Appendix Figure 12: Model Trace*


Design | Response Results | Pivot Grid | Reports | Dashboard Reports | Input Analysis

| Scenario | | | Replications | | Controls | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Name | Status | Required | Completed | BackstageCap | JewelryCap | WomensCap | SnJCap | PerfumeCap | MenChildCap | FnACap |
| ▸ | Scenario 1 | Comple... | 10 | 10 of 10 | 2 | 2 | 2 | 4 | 2 | 2 | 4 |
| * | | | | | | | | | | | |

*Appendix Figure 13: Base Model Controls*



| Design | Response Results | Pivot Grid | Reports | Dashboard Reports | Input Analysis |
|---|---|---|---|---|---|

| Scenario | | | Replications | | Controls | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Name | Status | Required | Completed | BackstageCap | JewelryCap | WomensCap | SnJCap | PerfumeCap | MenChildCap | FnACap |
| ▸ | Scenario 1 | Comple... | 10 | 10 of 10 | 4 | 3 | 5 | 7 | 3 | 4 | 9 |
| * | | | | | | | | | | | |

*Appendix Figure 14: Capacity Alternative Controls*

| Replication | Base W2 | Cap. Alt W2 | Const. Alt W2 | Difference of Cap Alt and Base | Difference of Const Alt and Base |
|---|---|---|---|---|---|
| 1 | 8.092358742 | 1.923052139 | 4.258981308 | -6.169306603 | -3.833377435 |
| 2 | 7.869790083 | 1.835275896 | 4.114253999 | -6.034514187 | -3.755536084 |
| 3 | 7.574956912 | 1.873680685 | 4.060400205 | -5.701276227 | -3.514556707 |
| 4 | 7.784760288 | 1.859009944 | 3.933034199 | -5.925750343 | -3.851726089 |
| 5 | 7.867853626 | 1.875225338 | 4.059494281 | -5.992628288 | -3.808359345 |
| 6 | 7.870494607 | 1.775157016 | 4.08069918 | -6.095337591 | -3.789795427 |
| 7 | 7.340096981 | 1.92198255 | 3.94409101 | -5.418114431 | -3.396005971 |
| 8 | 7.655791832 | 1.93896967 | 4.310458282 | -5.716822161 | -3.345333549 |
| 9 | 7.774091625 | 1.945758496 | 4.461546782 | -5.828333129 | -3.312544843 |
| 10 | 7.477762403 | 1.797516054 | 4.136347893 | -5.680246349 | -3.34141451 |
| AVG | | | | -5.856232931 | -3.594864996 |
| StDev | | | | 0.230350258 | 0.232010729 |

| Standard Error (Standard Deviation / Sqrt(n)) | | |
|---|---|---|
| Cap Alt | 0.072843 | |
| Const Alt | 0.073368 | |
| **Confidence Interval (Avg D +/- SE)** | | |
| | Upper CI | Lower CI |
| Cap Alt | -5.78339 | -5.92908 |
| Const Alt | -3.5215 | -3.66823 |

*Appendix Figure 15: Paired T-Test on Replications for W2*

| Replication | Base M1 | Cap. Alt M1 | Const. Alt M1 | Difference of Cap Alt and Base | Difference of Const Alt and Base |
|---|---|---|---|---|---|
| 1 | 3.426567996 | 1.065564798 | 1.012376276 | -2.361003197 | -2.414191719 |
| 2 | 3.683880152 | 0.90217354 | 0.983289352 | -2.781706612 | -2.7005908 |
| 3 | 3.39457923 | 0.978932045 | 1.138349584 | -2.415647185 | -2.256229645 |
| 4 | 3.304143973 | 0.94255666 | 1.12122574 | -2.361587314 | -2.182918233 |
| 5 | 3.625216404 | 0.980735763 | 1.326965253 | -2.644480641 | -2.298251151 |
| 6 | 3.368179717 | 0.927102942 | 1.123931171 | -2.441076775 | -2.244248545 |
| 7 | 3.278658709 | 0.97170758 | 0.917026908 | -2.306951129 | -2.361631801 |
| 8 | 3.404947991 | 1.017291311 | 1.18945628 | -2.387656681 | -2.215491711 |
| 9 | 3.309426898 | 1.004025572 | 0.811249579 | -2.305401326 | -2.49817732 |
| 10 | 3.153605706 | 0.915938684 | 1.087169484 | -2.237667022 | -2.066436222 |
| AVG | | | | -2.424317788 | -2.323816715 |
| StDev | | | | 0.166198325 | 0.179920373 |

| Standard Error (Standard Deviation / Sqrt(n)) | | |
|---|---|---|
| Cap Alt | 0.052557 | |
| Const Alt | 0.056896 | |
| **Confidence Interval (Avg D +/- SE)** | | |
| | Upper CI | Lower CI |
| Cap Alt | -2.37176 | -2.47687 |
| Const Alt | -2.26692 | -2.38071 |

*Appendix Figure 16: Paired T-Test on Replications for M1*

| Replication | Base M2 | Cap. Alt M2 | Const. Alt M2 | Difference of Cap Alt and Base | Difference of Const Alt and Base |
|---|---|---|---|---|---|
| 1 | 6.903344126 | 1.492012639 | 2.280562104 | -5.411331487 | -4.622782022 |
| 2 | 6.718850108 | 1.521554129 | 2.166456369 | -5.197295979 | -4.552393739 |
| 3 | 6.772748111 | 1.534907517 | 2.241797244 | -5.237840594 | -4.530950867 |
| 4 | 6.820136563 | 1.535905801 | 2.12015669 | -5.284230763 | -4.699979873 |
| 5 | 6.889359219 | 1.435399968 | 2.163574156 | -5.453959251 | -4.725785063 |
| 6 | 7.134263799 | 1.41675708 | 2.188447701 | -5.717506719 | -4.945816098 |
| 7 | 6.776542766 | 1.458264416 | 2.153182278 | -5.318278349 | -4.623360487 |
| 8 | 6.836785264 | 1.551608113 | 2.227615021 | -5.285177151 | -4.609170243 |
| 9 | 6.687422828 | 1.463945651 | 2.325382786 | -5.223477178 | -4.362040042 |
| 10 | 7.037416566 | 1.463186063 | 2.301082029 | -5.574230503 | -4.736334537 |
| AVG | | | | -5.370332797 | -4.640861297 |
| StDev | | | | 0.169054721 | 0.153516263 |

| Standard Error (Standard Deviation / Sqrt(n)) | | |
|---|---|---|
| Cap Alt | 0.05346 | |
| Const Alt | 0.048546 | |
| **Confidence Interval (Avg D +/- SE)** | | |
| | Upper CI | Lower CI |
| Cap Alt | -5.31687 | -5.42379 |
| Const Alt | -4.59232 | -4.68941 |

*Appendix Figure 17: Paired T-Test on Replications for M2*